

Rui Ma



Gender: Male

Mail: ruima@pku.edu.cn

Education: Ph.D.

Nationality: China

Objective: Internship

Available: April 2026-2027

Research Interests: [Embodied Intelligent, AI Security, Multimodal Info. Processing](#)

Google Scholar: <https://scholar.google.com.hk/citations?user=IJTYZIUAAAAJ&hl=en>

➤ Education

Sep 2020 -Jan 2025

Peking University

Computer Application Technology / Ph.D.

Advisors: Dr. Xiaodong Xie, Asst. Prof. Shanghang Zhang

Thesis: Blind Watermarking Algorithm Research for Image Copyright Protection

Sep 2017 -Jan 2020

University of Chinese Academy of Sciences

Optical Engineering / M.S.

Advisors: Prof. Yishi Shi

Thesis: Visible-Light Domain Ptychographic Microscopy and Application in Optical Information Hiding

Sep 2013 -Jan 2017

North China Electric Power University

Meas. & Control Tech./ B.E.

Thesis: Wind Power Prediction Based on Extreme Learning Machine Algorithm

➤ Experience

Jan 2026 -Present

Peking University

Postdoc

Co-advisor: Asst. Prof. Shanghang Zhang. **Research Interests:** Embodied intelligent safety;

Trustworthy multimodal large models; Copyright authentication; Traceability of generated content

July 2025 -Dec 2025

Ningxia University

Asst. Prof., M.S. Supervisor

Responsibilities: Research in computer vision, focusing on disinformation detection and security authentication; Teach CS courses; Supervise postgraduates in thesis completion and paper publication; Prepare and apply for research grants (e.g., National Natural Science Foundation, Provincial Foundation, Provincial Key R&D Programs)

Sep 2024 -May 2025

Yunsheng Intelligent Co., Ltd.

Intern. & AI Researcher

Responsibilities: Design and implement end-to-end autonomous driving solutions for quadcopter drones, integrating imitation & reinforcement learning strategies under the VLA framework. Explored implementing autonomous obstacle avoidance and navigation algorithms from RGB four-eye fisheye images to PVA control signals (edge side)

►► List of Publications

1. **R. Ma**, Y. Hou, C. Li, H. Jia, X. Xie, "Scene-Adaptive Unsupervised Crowd Counting for Video Surveillance" **IEEE TCSVT**, 2025, vol. 35, no. 7, pp. 6910-6925
2. **R. Ma**, S. Zhang, T. Ruan, Y. Tao, Y. Shi, "Scanning-position error-correction algorithm in dual-wavelength ptychographic microscopy" **Chinese Physics Letters**, 2020, 37(4)
3. **R. Ma**, Y. Li, H. Jia, Y. Shi, X. Xie, T. Huang, "Optical information hiding with phase-only and non-mechanical ptychography" **Optics & Lasers in Engineering**, 2021, 141, 106569
4. **R. Ma**, D. Yang, T. Yu, T. Li, Y. Shi, "Sharpness-statistics-based auto-focusing algorithm for optical ptychography" **Optics & Lasers in Engineering**, 2020, 128, 106053
5. Y. Hou, S. Zhang, **R. Ma**, H. Jia, X. Xie, "Frame-Recurrent Video Crowd Counting" **IEEE TCSVT**, 2023, 33 (9), 5186-5199
6. X. Sun, S. Zhang, **R. Ma**, Y. Tao, Y. Zhu, D. Yang, Y. Shi, "Natural speckle-based watermarking with random-like illuminated decoding" **Optics Express**, 2020, 28 (21)
7. J. Zhang, D. Yang, **R. Ma**, Y. Shi, "Multi-image and color image encryption via multi-slice ptychographic encoding" **Optics Communications**, 2021, 485, 126762
8. Y. Tao, D. Yang, **R. Ma**, Y. Zhu, Y. Shi, "Enhanced-visual-cryptography-based optical information hiding system" **Acta Physica Sinica**, 2020, 69 (14)
9. D. Miao, **R. Ma**, X. Sun, Y. Shi, "Chosen-plaintext attack to linear unitary optical encryption systems" **Journal of University of Chinese Academy of Sciences**, 2022, 39 (2), 165

►► Conference

1. **R. Ma**, M. Guo, Y. Hou, F. Yang, Y. Li, H. Jia, X. Xie, "Towards Blind Watermarking: Combining Invertible and Non-invertible Mechanisms" **ACM Multimedia**, 2022. [Citations: 150+]
2. **R. Ma**, M. Guo, P. Jia, C. Li, Y. Hou, Y. Li, X. Xie, S. Zhang, "Enhanced Blind Watermarking Against Black-Box Noise: Leveraging CIN Framework" **IEEE ICME (Oral)**, 2024
3. M. Gong, **R. Ma**, H. Liu, "WaveFace: Haar Structural Alignment and Shuffling for Privacy-Preserving Face Recognition" **ISCAIT**, 2026

►► Submitted

1. **R. Ma**, Q. Zhou, S. Zhou, Y. Jin, D. Zhou, B. Xiao, X. Li, Y. Qu, A. Singh, K. Keutzer, J. Hu, X. Xie, Z. Dong, S. Zhang, "A Dataset and Benchmark for Copyright Infringement Unlearning from Text-to-Image Diffusion Models" (Under Review in **IJCV**)
2. **R. Ma**, M. Guo, Y. L, H. Zhang, C. Ma, Y. Li, X. Xie, S. Zhang, "PiGW: A Plug-in Generative Watermarking Framework" (Under Review in **IEEE TMM**)

3. **R. Ma**, B. Sun, M. Gong, Z. Li, Y. Wu, H. Liu "ALBS: Adaptive Latent Basis Shrinkage for Generalizable Deepfake Detection" (**Under Review in TIFS**)

►► Patents

1. CN: CN112132747A, 2020-12-25
2. CN: CN111563938A, 2020-08-21
3. CN: CN111526261A, 2020-08-11
4. CN: CN111383292A, 2020-07-07
5. CN: CN111382449A, 2020-07-07
6. CN: CN111383291A, 2020-07-07
7. CN: CN111385435A, 2020-07-07
8. CN: CN111375884A, 2020-07-07
9. CN: CN111383290A, 2020-07-07
10. CN: CN111241554A, 2020-06-05
11. CN: CN110942413A, 2020-03-31
12. CN: CN110728164A, 2020-01-24
13. CN: CN110360924A, 2019-10-22
14. CN: CN110308117A, 2019-10-08

►► Awards and Honors

1. Natural Science Foundation of Ningxia for **Outstanding Youth** (Class B) – 2026
2. China College Students' "Internet+" Innovation and Entrepreneurship Competition - Provincial **Gold Award** – 2024
3. Peking University - NERC Annual **Outstanding Student** – 2021
4. University of Chinese Academy of Sciences - **Merit Student** – 2020
5. University of Chinese Academy of Sciences - Sec. Class **Scholarship** - 2018 / 2019 / 2020
6. National High School Physics Competition (**CPhO**) - Ningxia Division First Prize – 2013

►► Review for Journals & Conference

Conference on Neural Information Processing Systems (**NeurIPS**); IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**); ACM International Conference on Multimedia (**ACM MM**); IEEE International Conference on Multimedia and Expo (**ICME**); Pattern Recognition and Computer Vision Conference (**PRCV**); Chinese Conference on Biometric Recognition (**CCBR**); IEEE Transactions on Circuits and Systems for Video Technology (**IEEE TCSVT**); IEEE Transactions on Multimedia (**IEEE TMM**); IEEE Transactions on Dependable and Secure Computing (**IEEE TDSC**); Optics & Lasers in Engineering (**Opt. Lasers Eng.**).

►► Memberships

- Association for Computing Machinery (**ACM Member**)
- Institute of Electrical and Electronics Engineers (**IEEE Member**)
- China Computer Federation (**CCF Member**)
- China Society of Image and Graphics (**CSIG Member**)

►► Selected Grants

- **Natural Science Foundation of Ningxia for Outstanding Youth (Class B)| Principal Investigator|2026.**
Project: A Generative Blind Watermarking Framework for Generative Models
- **Ministry of Industry and Information Technology (MIIT) High-Quality Project | Participant| 2022 – 2025**
Project: AI Software and Hardware Adaptation & Migration Technology and Tools
- **CCF-Baidu Pinecone Fund | Participant| 2022 – 2023**
Project: Cross-Scenario Generalization for 3D Perception in Autonomous Driving